GREEN AI: Making AI More Sustainable

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AI's Unnoticed Environmental Impact

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AI has been rapidly changing our world and how we interact with it. Whether it is scientists predicting advanced protein folds, small businesses deploying AI websites with AWS, or customer service applying chatbots for customer interaction, nearly everyone is interacting with artificial intelligence in some form. Machine learning has become one of the most popular specializations for computer science developers.

Despite AI's international ubiquity among hundreds of millions of users, most are unfortunately unfamiliar with terms like "Green AI" or "carbon-aware computing." Sustainability rarely surfaces in discussions about software architecture or model development even though incorporating sustainable practices into software engineering is crucial, especially looking at the environmental cost of these large GenAI models.

Running these massive Generative AI models requires a significant amount of energy. AI queries require ten times the energy of a regular Google search. Training the BLOOM Generative AI model resulted in 50 tonnes of greenhouse gas emissions total. OpenAi's training of GPT-3, one of the smaller models, produced 502 total tonnes of emissions. More modern, complex models required even more. As our models grow, the pollution required to train and run these algorithms will balloon in size; it is projected that 30% of total worldwide energy usage will be from AI. Not only do running and developing these models require exorbitant amounts of pollution to be released, but a large amount of clean water is required to properly cool and maintain the hyperactive GPU's and TPU's of the machines running these models. As more people use AI and more complicated models are being trained, the cost to the environment and our resources will grow in magnitudes.

The term "Green AI" connects two key aspects: sustainability and artificial intelligence. On one hand, it focuses on how AI can be applied to improve sustainable practices. Examples include developing predictive models for environmental change, forecasting water usage in agriculture, and supporting wildlife conservation efforts.

On the other hand, it looks at how to make the training and development of AI models themselves more sustainable. This includes monitoring carbon footprints, optimizing hyperparameter tuning, and designing more efficient algorithms.

This paper will highlight the sustainability costs of AI models and explore ways to incentivize environmentally conscientious choices, making terms like Green AI and carbon-aware computing more

prominent in the Big Data community, and leading to actionable outcomes that promote environmental responsibility.

Sustainable AI Practices and Solutions

As AI has such a pronounced environmental impact, it's important to find strategies to reduce this footprint. This, of course, also means that both companies and AI users need to prioritize environmental responsibility; methods for this will be explored in later sections. But before that, it is useful to examine some general approaches to reducing the environmental impact of AI systems.

For example, many large-scale companies deploy AI models on AWS or Microsoft Azure, both of which operate data centers across multiple regions worldwide. Each data center has a different Power Usage Effectiveness (PUE) value due to variations in cooling efficiency, climate, and infrastructure. For example, electricity in North Central Sweden is largely generated from low-carbon, renewable sources, making data centers in this region more sustainable to operate. In contrast, data centers located in warmer or less energy-efficient regions require significantly more energy for cooling, increasing their overall environmental impact. A PUE closer to 1 indicates greater energy efficiency; for instance, the BitFury data center in Georgia has a PUE of approximately 1.02, while Amazon's most efficient regions, such as those in Europe, have reported PUE values around 1.04. Being intentional about which data centers are selected from a sustainability perspective and incentivizing this type of decision-making would be an ideal first step toward making AI deployments more carbon-efficient.

Additionally, hardware optimization techniques can be employed. One option could be opting for specialized energy-efficient process units with reduced-complexity instruction sets made specifically for machine learning operations, which will minimize the instruction translation overhead. Some examples are Tensor Processing Units (TPUs), Neural Processing Units (NPUs), or Graphics Processing Units (GPUs) with Tensor Cores.

Another important factor in improving AI efficiency is the use of smaller, task-appropriate models instead of heavily overparameterized architectures, which often incur substantial energy costs while providing only marginal gains in performance. For example, many classification or information extraction tasks can be effectively handled by fine-tuned lightweight transformer models such as DistilBERT or ALBERT, rather than large-scale models with billions of parameters, achieving comparable accuracy while requiring significantly less compute and energy during both training and inference. Expanding on this, when using a virtual server to deploy models in the cloud, selecting instance types that are scaled for the necessary workload, rather than overproviding with a high-performing instance type, can further reduce unnecessary energy consumption.

Gamifying Sustainability for Awareness and Adoption

Increasing AI users' awareness of their environmental impact, guiding more efficient AI usage, and actively incentivizing the implementation of sustainable techniques will help businesses and users prioritize sustainability, rather than treating

Defining for users their specific environmental impact, providing tailor-made strategies to further reduce carbon usage, and actively incentivizing the implementation of sustainable techniques will help businesses and users prioritize sustainability, rather than treating the environmental effects as an afterthought, if they are considered at all.

The first step is improving monitoring techniques. LLM API usage does provide information on token usage, but it would be helpful to also have LLM APIs return environmental impact data as part of their responses, alongside standardized calculations of carbon emissions, water usage, and electricity consumption across model infrastructure. By attaching quantifiable metrics to these impacts, users gain a concrete understanding of their exact footprint, replacing vague or abstract guilt with actionable, measurable insight.

Of course, simply looking at the carbon footprint of AI usage on its own can be misleading, since different users rely on AI for very different purposes. For example, one user might use AI to better understand lecture notes but do so inefficiently—by inputting excessive text to summarize or repeatedly re-asking similar questions. While this may reflect inefficient energy use, the overall carbon footprint is still relatively small compared to someone developing a sophisticated AI model to predict potential healthcare treatment side effects before a drug enters clinical trials.

What matters is not just measuring carbon consumption, but using that data to identify areas where efficiency can be improved. By highlighting these inefficiencies, we can incentivize users to adopt more sustainable and energy-conscious practices.

There are a few ways the user can be presented with how they are misusing their resources, which produces unnecessary emissions. For example, imagine an AWS dashboard that highlights underutilized resources, like an EC2 instance that isn't being used effectively. By right-sizing or downsizing that instance, the user could gain sustainability points, rewarding optimization, and encouraging more efficient resource usage. Also, points can be accrued by using more sustainable hardware that is known for efficiency when it comes to large ML workloads.

Users earn points that function like a credit reward system, similar to a cashback credit card. These points can be redeemed for billing credits toward AI usage costs or even converted into monetary cashback. In addition, users, projects, and companies can receive awards and badges for sustainable

practices, representing recognition for implementing efficient, low-impact AI workflows. These badges signal employees and teams with strong knowledge of responsible AI operations.

Now for clarification, in rewarding users in sustainability, we want to have a comparison of expected energy usage for the problem at hand, and then compare to the users actual usage of energy. This will require really indepth methods of calculating averages. Of course, studies could be done for different AI use cases to measure how much average users with similar use cases end up producing emissions-wise, there could also be AI platforms (like AWS) that incentivizes users to deploy and run AI models through their services, and within these services monitoring can be done for very similar use cases, and calculate the average expected emissions from such a use case.

By comparing projects based on their AI awareness awards, organizations can quickly identify high-performing, sustainability-mature initiatives and highlight weaker areas that still need development. On a broader scale, company-wide comparisons of environmental awareness indicate which organizations are leading in responsible AI adoption and which may require guidance, new policies, or technical improvements to reduce resource waste and operational emissions.

Case Studies and Benchmarks

Currently there are few products ready on the market that are built to gamify sustainability for AI users. Although there are a few initiatives that incorporate aspects of this goal that can be discussed.

Several tools have been developed to help monitor users' sustainability impacts. One example is AWS's Customer Carbon Footprint Tool (CCFT), which provides visibility into emissions associated with cloud usage.

These emissions are typically divided into two categories. The first is Scope 2, which refers to indirect emissions from the electricity used to power services such as EC2 instances and S3 storage. The second is Scope 3, which includes emissions associated with the manufacturing, transportation, installation, and end-of-life treatment of the hardware that supports those cloud services.

Together, these categories offer a more complete view of the environmental impact of using cloud infrastructure. Of course AWS's footprint tool has some major limitations, namely that it aggregates usage data rather than displaying specific workload data. But, the tool overall is a major step in the right direction, as it attempts to calculate the predicted emissions and the actual emissions consumed, and as AWS is used by many large scale companies already, these kinds of integrated tools can really aid in different user's and companies overall AI impact.

A look at the gamification aspect; several recycling “money back” types of structures have been implemented where recycling plastic has earned back profits on households to try to encourage users to throw plastic in recycling bins rather than trash bins. In Singapore there are essentially reverse vending machines, specifically made to pay back for deposition plastic bottles.

Other similar systems exist called Deposit-return schemes; these programs essentially have a deposit in purchasing drinks that can be returned after the empty container is returned. Some countries that have this system implemented include Germany, Norway, and the United States, and these systems lead to large scale participation with bottle return rates of 80-90%

A major avenue for building up functioning Sustainability and AI is through startups.

There are several AI startup initiatives that are helping to bring more awareness to a user’s specific carbon contributions. For an example. CO2 AI attempts to automate tracking and analyzing carbon emissions for companies. There are several other startups with similar goals in emissions tracking like Watershed, Climatiq, Klima.

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Policy

How AI can help the environment - Google Earth AI,

Negatives of No AI

Policy - having governments require measurement and reporting of environmental impacts

Added Benefits

Locally usable models for the vast majority of non-power users is optimal. Saves on datacenter resource usage. If it is locally run there are fewer privacy concerns as well. System could also be something where the application that allows the model to be run locally prioritizes local use but if there is need for cloud compute it automatically and securely send that over. Find figures showing what percent of queries are actually doable on average user hardware. Also figure out total number of AI queries per year. Get a find OMG number based on that.

Also for corporations if they invest in the research, development, an deployment of optimized and more efficient hardware and software they get to save money on costs associated with resource use be it water, energy, personnel required for datacenter maintenance etc. Find graphs showing cost decreases for datacenter management as time goes on.

Call to Action and Final Thoughts

With the bloom of AI usage, sustainability should be incorporated into the strategy governing how models are trained, deployed, and used. However, the focus in AI development is, at times, placed on improving metrics such as accuracy, throughput, and benchmark performance, to the extent that sustainability-related considerations are often overlooked or not widely understood. Increasing awareness of Green AI, alongside the development of systems and frameworks that transparently quantify the environmental impact of AI usage, can help users better understand their individual contribution and the broader implications of AI at scale. By making these impacts more visible, sustainability can be elevated to a core priority rather than an afterthought.